

Deep Learning Technical Report

Skill Set Go EduTech · AI/ML Track · Week 2

1. Objective

This report compares the performance of a feed-forward neural network and a convolutional neural network (CNN) trained in PyTorch on the scikit-learn digits dataset (1,797 8x8 grayscale images of handwritten digits, 0-9), and summarizes the outcome of a hyperparameter experimentation run across different optimizers, learning rates, and batch sizes.

2. Model Comparison

Both models were evaluated on the same 20% held-out test split (360 samples).

Model	Architecture	Test Accuracy	Notes
Feed-forward NN	64 → 32 → 32 → 10, ReLU	96.39%	Trained on flattened pixel vectors, 10 epochs
CNN	2 conv layers + pooling + FC	96.39%	Trained on raw 8×8 images, 15 epochs, preserves spatial structure

Both models landed at exactly the same test accuracy (96.39%) on this dataset. This is a small, fairly easy 8×8 image classification problem, so a well-tuned feed-forward network has enough capacity to match a CNN here — the CNN's structural advantage (preserving spatial relationships between neighboring pixels) tends to matter more as image size and complexity grow. On larger, more complex images the gap between the two architectures would be expected to widen in the CNN's favor.

3. Hyperparameter Experiment Summary

Five experiments were run on the feed-forward network, varying optimizer, learning rate, batch size, and epoch count.

Optimizer	LR	Batch Size	Epochs	Accuracy	Train Time (s)
SGD	0.01	16	10	80.28%	2.46
SGD	0.1	16	10	97.78%	2.11
Adam	0.001	16	10	97.78%	3.39
Adam	0.001	64	10	95.83%	1.09
Adam	0.001	16	20	97.50%	6.22

4. Findings

- SGD with a low learning rate (0.01) badly underperformed everything else, reaching only 80.28% accuracy versus 95–98% for every other configuration — at this learning rate, 10 epochs simply wasn't enough for SGD to converge.
- Raising the SGD learning rate to 0.1 closed the gap completely, matching Adam's 97.78% at the same batch size and epoch count — proving the earlier gap was a tuning issue, not a fundamental weakness of SGD.
- Adam with a larger batch size (64) trained nearly 3× faster per run (1.09s vs 3.39s) but lost about 2 points of accuracy (95.83% vs 97.78%), since fewer, larger batches mean fewer weight updates overall.
- Doubling Adam's epochs from 10 to 20 did not improve accuracy (97.78% → 97.50%) and nearly doubled training time, suggesting the model had already converged by epoch 10 and started slightly overfitting afterward.
- The best-tuned feed-forward configuration (97.78%, Adam or SGD @ 0.1) actually edged out the specific run logged for the model comparison above (96.39%) — the take-away is that hyperparameter choice can matter as much as architecture choice on a small, simple dataset like this one.

5. Conclusion

Adam is the safer default optimizer: it reached strong accuracy without any learning rate tuning, while SGD needed its learning rate raised 10× to be competitive. The best cost-to-accuracy tradeoff in this experiment set was Adam at a small batch size (16) and 10 epochs — larger batches traded accuracy for speed, and more epochs added training time without adding accuracy. On this particular dataset the feed-forward network matched the CNN's accuracy exactly, but the CNN's architecture-level advantage in capturing spatial structure would be expected to matter more on larger, more complex images than these 8×8 digit scans.